

II-Probabilistic Approaches Independence and Graphical Models

Foundations of Neuro-Symbolic AI

Alex Goessmann

University of Applied Science Würzburg-Schweinfurt (THWS)

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Outline

- ▶ Independence of variables
- ▶ Conditional independence
- ▶ Decomposition schemes
- ▶ Bayesian networks
- ▶ Markov networks

How to beat the curse of dimensions?

Probability distribution of factored systems with states $\times_{k \in [d]} [m_k]$ has

$$\left(\prod_{k \in [d]} m_k \right) - 1$$

degrees of freedom (coordinates to specify and store).

Mitigation: [Tensor network decompositions](#)

Idea

Exploit the "dependence" structure of variables to find decompositions of probability distributions.

Example: At the dentist

Consider the joint distribution of the variables X_0 (Toothache), X_1 (Cavity), X_2 (Catch), with a variable X_3 , denoting whether there is a Cloud outside the dentists lab.

- ▶ We now have an order-4 probability tensor $\mathbb{P}[X_0, X_1, X_2, X_3]$
- ▶ Intuitively, knowing Cloud should not affect the probability of having a cavity, so why should we care?

To be more precise let us consider the distribution:

$$\mathbb{P}[X_0, X_1, X_2, X_3] = \begin{pmatrix} 0.288 & 0.004 \\ 0.032 & 0.006 \end{pmatrix} \begin{pmatrix} 0.077 & 0.036 \\ 0.008 & 0.054 \end{pmatrix} [X_0, X_1, X_2, X_3]$$

Such redundancies are captured by the concept of independence.

Formal definition of independence

Definition (Independence)

Given a joint distribution $\mathbb{P}[X_{\mathcal{V}}]$ and two disjoint subsets $A, B \subset \mathcal{V}$, we say that the variables X_A are independent from the variables X_B if the marginal distributions $\mathbb{P}[X_{A \cup B}]$, $\mathbb{P}[X_A]$ and $\mathbb{P}[X_B]$ satisfy for any index tuples x_A, x_B

$$\mathbb{P}[X_A = x_A, X_B = x_B] = \mathbb{P}[X_A = x_A] \cdot \mathbb{P}[X_B = x_B] .$$

We denote in this case $(X_A \perp X_B)$.

Interpretation: Knowing the value of variables B (and conditioning on them) does not change the distribution of variables A .

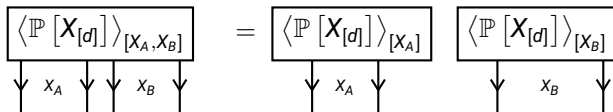
Contraction criterion for independence

Theorem

If and only if the X_A are independent from the variables X_B with respect to a distribution $\mathbb{P}[X_V]$ then:

$$\langle \mathbb{P}[X_{[d]}] \rangle_{[X_A, X_B]} = \langle \mathbb{P}[X_{[d]}] \rangle_{[X_A]} \otimes \langle \mathbb{P}[X_{[d]}] \rangle_{[X_B]} .$$

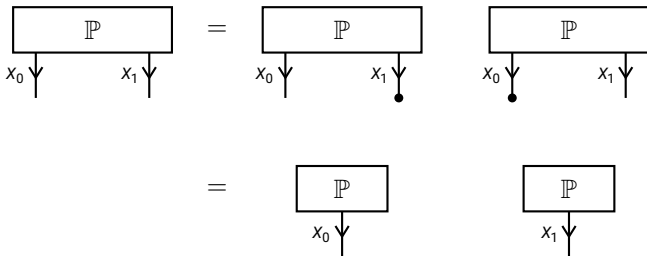
In a tensor network diagram this is the decomposition:



Decomposition into marginal distributions

Independence allows the decomposition of probability tensors into tensor products.

Example: The distribution of two independent variables decomposes as:



Storage advantage

Instead of storing $m_0 \cdot m_1$ coordinates, we can store $\mathbb{P}$ in the case of independence with $m_0 + m_1$ demand.

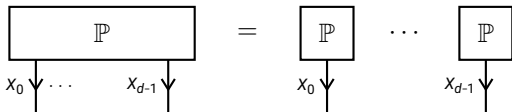
Decomposition of pairwise independent variables

Imagine now that we have a distribution $\mathbb{P} [X_{[d]}]$, such that for any $k, \tilde{k} \in [d]$

$$(X_k \perp X_{\tilde{k}}) .$$

We then have the decomposition:

$$\mathbb{P} [X_{[d]}] = \bigotimes_{k \in [d]} \mathbb{P} [X_k]$$



Exponential to linear storage demand

Instead of storing $\prod_{k \in [d]} m_k$ coordinates, we can store $\mathbb{P}$ using $\sum_{k \in [d]} m_k$ coordinates of the marginal distributions.

Outline

- ▶ Independence of variables
- ▶ **Conditional independence**
- ▶ Decomposition schemes
- ▶ Bayesian networks
- ▶ Markov networks

Assumption of independence

Independence is a very restrictive assumption:

- ▶ We need dependence of effects on causes for machine learning to work.
- ▶ Independence covers only the most extreme form of tensor network decompositions.

Example: In the dentist example (without the **Cloud** variable), no variables are independent from each other.

Are there more common properties corresponding to more generic tensor decompositions?

Formal definition of Conditional Independencies

Definition (Conditional Independence)

Given a joint distribution of variables X_A , X_B and X_C , we say that X_A and X_B are independent given X_C if for any index tuple (x_A, x_B, x_C)

$$\mathbb{P}[X_A = x_A, X_B = x_B | X_C = x_C] = \mathbb{P}[X_A = x_A | X_C = x_C] \cdot \mathbb{P}[X_B = x_B | X_C = x_C],$$

We denote this situation by

$$(X_A \perp X_B) \mid X_C.$$

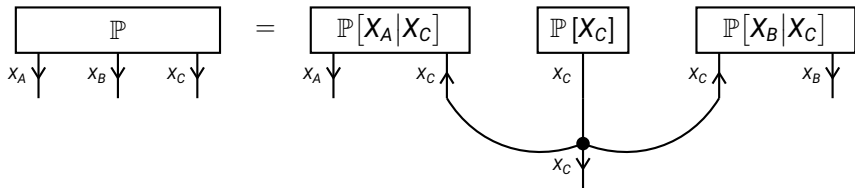
Interpretation: Conditional independence means that X_A and X_B are independent with respect to any slice $\mathbb{P}[X_A, X_B | X_C = x_C]$.

Conditional independence as a contraction equation

We have $(X_A \perp X_B) \mid X_C$ if and only if

$$\mathbb{P}[X_A, X_B \mid X_C] = \langle \mathbb{P}[X_A \mid X_C], \mathbb{P}[X_B \mid X_C] \rangle_{[X_A, X_B, X_C]} .$$

We depict conditional independence by tensor network decompositions:



Example: Being at a dentist

Task: Let us investigate, whether **Toothache** (X_0) and **Catch** (X_2) are independent given **Cavity** (X_1), with respect to the distribution:

$$\mathbb{P}[X_0, X_1, X_2] = \begin{pmatrix} 0.576 & 0.008 \\ 0.064 & 0.012 \end{pmatrix} \begin{pmatrix} 0.144 & 0.072 \\ 0.016 & 0.108 \end{pmatrix} [X_0, X_1, X_2]$$

For $X_1 = 0$:

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$$\mathbb{P}[X_0, X_2 | X_1 = 0] = \begin{pmatrix} 0.72 & 0.08 \\ 0.18 & 0.02 \end{pmatrix} [X_0, X_2]$$

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For $X_1 = 1$:

$$\mathbb{P}[X_0, X_2 | X_1 = 1] = \begin{pmatrix} 0.04 & 0.36 \\ 0.06 & 0.54 \end{pmatrix} [X_0, X_2]$$

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Example: Being at a dentist

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Conclusion: The conditional independence holds, that is $(X_0 \perp X_2) | X_1$.

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- ▶ Independence of variables
- ▶ Conditional independence
- ▶ **Decomposition schemes**
- ▶ Bayesian networks
- ▶ Markov networks

Chain Rule: Decomposing Probabilities

Theorem (Chain Rule)

For any joint probability distribution $\mathbb{P}[X_{[d]}]$ we have

$$\mathbb{P}[X_{[d]}] = \langle \{\mathbb{P}[X_k | X_{[k]}] : k \in [d]\} \rangle_{[X_{[d]}]} .$$

Proof sketch: We apply the Bayes theorem to get

$$\mathbb{P}[X_{[d]}] = \langle \mathbb{P}[X_0], \mathbb{P}[X_1, \dots, X_{d-1} | X_0] \rangle_{[X_{[d]}]} .$$

and further apply the Bayes theorem iteratively on the conditioned distribution.

Problem: The tensors $\mathbb{P}[X_k | X_{[k]}]$ are of order $d + 1$!

Can we reduce the order based on independence assumptions?

Simplification of conditional distributions

Theorem

When $(X_A \perp X_B) \mid X_C$ then

$$\mathbb{P}[X_A|X_C, X_B] = \mathbb{P}[X_A|X_C] \otimes \mathbb{I}[X_B] .$$

Proof sketch: We have by Bayes theorem

$$\mathbb{P}[X_A, X_B|X_C] = \langle \mathbb{P}[X_A|X_C, X_B], \mathbb{P}[X_B|X_C] \rangle_{[X_A, X_B, X_C]}$$

and compare with the conditional independence equation

$$\mathbb{P}[X_A, X_B|X_C] = \langle \mathbb{P}[X_A|X_C], \mathbb{P}[X_B|X_C] \rangle_{[X_A, X_B, X_C]} .$$

Usage for decompositions: We can drop some conditioned variables and effectively reduce the tensor order in the tensors $\mathbb{P}[X_k|X_{[k]}]$.

Example: Markov Chains

Let us model the weather in a town on all days of a year by variables $X_{[365]}$ and assume that for each $k \in [365]$

- ▶ Three states $x_k = 0$ (sunny), $x_k = 1$ (cloudy), $x_k = 2$ (rainy)
- ▶ The weather on the day $k + 1$ conditioned on the weather on the day k is

$$\mathbb{P}[X_{k+1}|X_k] = \begin{pmatrix} 0.70 & 0.25 & 0.05 \\ 0.35 & 0.40 & 0.25 \\ 0.10 & 0.40 & 0.50 \end{pmatrix} [X_{k+1}, X_k]$$

- ▶ The weather on the day $k + 1$ does not depend on the weather before day k given the weather before day k :

$$(X_{k+1} \perp X_{[k]}) \mid X_k$$

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- ▶ The weather on the day $k + 1$ does not depend on the weather before day k given the weather before day k :

$$(X_{k+1} \perp X_{[k]}) \mid X_k$$

This is an example of a Markov chain: Variables depend only on their predecessors.

Example: Markov Chains

Theorem (Markov Chain)

Let there be a set of variables X_k where $k \in [d]$. When X_k is independent of $X_{[k-1]}$ given X_{k-1} (the Markov Property), then

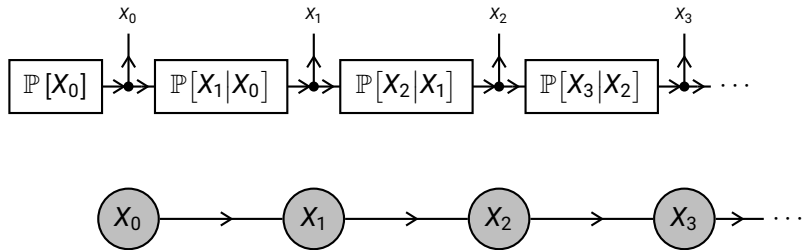
$$\mathbb{P}[X_{[d]}] = \langle \mathbb{P}[X_k | X_{k-1}] : k \in [d] \rangle_{[X_{[d]}]} .$$

Proof sketch: We use the chain decomposition and exploit the independence assumptions to get

$$\mathbb{P}[X_k | X_{[k]}] = \mathbb{P}[X_k | X_{k-1}] \otimes \mathbb{I}[X_{[k-1]}] .$$

Tensor network decompositions of markov chains

We depict this by



Can we find a more generic correspondence between probability decompositions and conditional independence statements?

Outline

- ▶ Independence of variables
- ▶ Conditional independence
- ▶ Decomposition schemes
- ▶ **Bayesian networks**
- ▶ Markov networks

Independence assumptions of Bayesian networks

Towards decomposing a distribution in a more generic way take the steps:

- ▶ We choose an order and enumerate the variables by $[d]$.
- ▶ To each $k \in [d]$ we choose a set $\text{Pa}(k)$ of **parent nodes** in $[k] = \{0, \dots, k-1\}$ such that

$$(X_k \perp X_{[k] \setminus \text{Pa}(k)}) \mid X_{\text{Pa}(k)}$$

Example: Markov chains are the special case of $\text{Pa}(k) = \{k-1\}$.

Factorization into Bayesian networks

Theorem

When for each $k \in [d]$

$$(X_k \perp X_{[k] \setminus \text{Pa}(k)}) \mid X_{\text{Pa}(k)}$$

then

$$\mathbb{P}[X_{[d]}] = \langle \{\mathbb{P}[X_k | X_{\text{Pa}(k)}] : k \in [d]\} \rangle_{[X_{[d]}]} .$$

Proof sketch: As in the proof of the Markov chain factorization theorem, we use the chain factorization and exploit the conditional independences to show that

$$\mathbb{P}[X_k | X_{[k]}] = \mathbb{P}[X_k | X_{\text{Pa}(k)}] \otimes \mathbb{I}[X_{[k] \setminus \text{Pa}(k)}] .$$

Example: At the Dentist

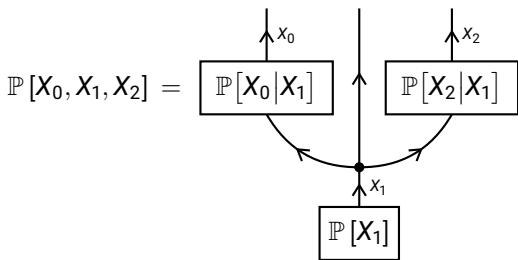
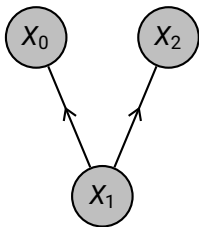
- Choose the order X_1, X_0, X_2
- For both X_0 and X_2 we take X_1 as single parent, that is

$$\text{Pa}(0) = \{1\} \quad \text{and} \quad \text{Pa}(2) = \{1\}$$

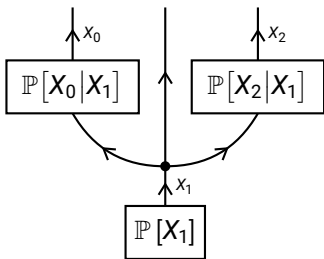
and have the for the Bayesian network necessary conditional independencies (shown before)

$$(X_0 \perp X_2) \mid X_1 \quad \text{and} \quad (X_2 \perp X_0) \mid X_1$$

The corresponding hypergraph is:



$$\mathbb{P}[X_0, X_1, X_2] =$$



Example: Hidden Markov Models

Extend the weather example: We observe the weather X_k , but not the hidden driving variable:

- ▶ Atmospheric pressure Y_k taking states $y_k = 0$ (high pressure) or $y_k = 1$ (low pressure)
- ▶ The pressure at a day $k + 1$ is independent of the pressure at days $[k]$ given the pressure at day k :

$$(Y_{k+1} \perp Y_{[k]}) \mid Y_k$$

- ▶ The weather at day k depends only on the pressure:

$$(X_k \perp Y_{[k]}, X_{[k]}) \mid Y_k$$

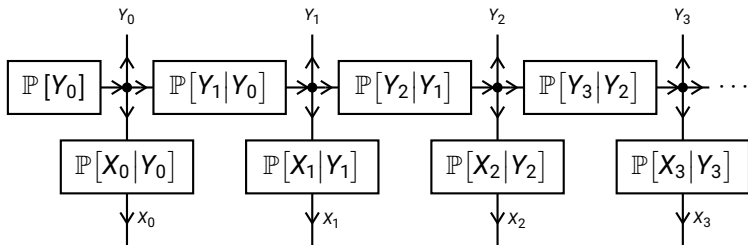
This is an example of Hidden Markov Models extending Markov Chains by limited observation X_k of the variables Y_k .

Example: Hidden Markov Models

The independence assumptions of a Hidden Markov Model are for any $k \in [d]$

- ▶ Y_{k+1} is independent of $Y_{[k]}$ and $X_{[k]}$ given Y_k
- ▶ X_k is independent of all other variables given Y_k

The independence assumptions are exploited in the decomposition



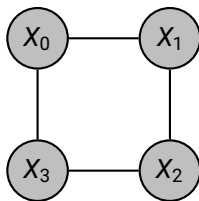
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Sometimes there is no clear order of the variables

Example: Consider four students enumerated by $k \in [4]$ and

- ▶ Boolean variables X_k whether they know the solution to a problem.
- ▶ They sit next to each other in a circle and sometimes pass the solution to their neighbor.



We have the independencies:

$$(X_0 \perp X_2) \mid X_1, X_3 \quad \text{and} \quad (X_1 \perp X_3) \mid X_0, X_2$$

But: Any order of the student variables in a Bayesian network would at most capture one of these two independencies.

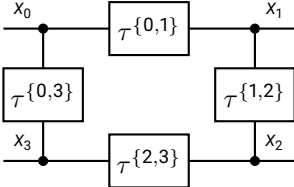
Representation as a Markov network

Let us make the neighboring students example more precise:

- ▶ For each pair of neighbors, we "increase" by a factor 2 the probability that both either know or do not know the solution.
- ▶ We define matrices:

$$\tau^{\{0,1\}}[X_0, X_1] = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} [X_0, X_1]$$

- ▶ We then model the distribution by a tensor network:

$$\mathbb{P}[X_{[4]}] = \frac{1}{\langle \tau^{\{0,1\}}, \tau^{\{1,2\}}, \tau^{\{2,3\}}, \tau^{\{0,3\}} \rangle_{[\emptyset]}} \cdot$$


Different to Bayesian networks: Need to **normalize** the distribution

$$\mathbb{P}[X_{[4]}] = \frac{1}{\langle \tau^{\{0,1\}}, \tau^{\{1,2\}}, \tau^{\{2,3\}}, \tau^{\{0,3\}} \rangle_{[X_{[4]}|\emptyset]}}$$

Undirected Graphical Models: Markov Networks

Definition (Markov Networks)

Let $\{\tau^e[X_e] : e \in \mathcal{E}\}$ be a Tensor Network on a hypergraph $\mathcal{G}$. The associated Markov Network is the probability distribution of $\{X_v : v \in \mathcal{V}\}$ defined by

$$\mathbb{P}[X_{\mathcal{V}}] = \frac{\langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{V}}]}}{\langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[\emptyset]}} = \langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[X_{\mathcal{V}}|\emptyset]} .$$

We call the denominator

$$\mathcal{Z}(\mathcal{G}) = \langle \{\tau^e[X_e] : e \in \mathcal{E}\} \rangle_{[\emptyset]}$$

the **partition function** of the Markov Network.

Markov network independencies

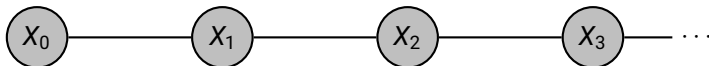
Which conditional independencies hold in a Markov network on a hypergraph $\mathcal{G}$?

Definition

For any hypergraph $\mathcal{G} = (\mathcal{E}, \mathcal{V})$ we define:

- ▶ A **path** through $\mathcal{G}$ is a selection of nodes $[d] \subset \mathcal{V}$ such that for any $k \in [d - 1]$ we find a hyperedge $e \in \mathcal{E}$ with $k, k + 1 \in e$.
- ▶ Given disjoint subsets $A, B, C \subset \mathcal{V}$ we say that C **separates** A from B if any path starting at a node in A and ending at a node in B goes through C .

Example: In a Markov chain graph, for any $k \in [d]$ any subset $A \subset [k]$ and subset $B \subset [d] \setminus [k + 1]$ is separated by $C = \{k\}$.



Markov network independencies

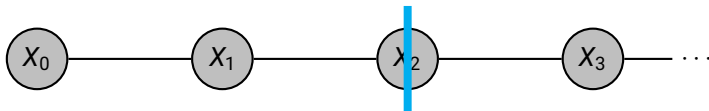
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Example: In a Markov chain graph, for any $k \in [d]$ any subset $A \subset [k]$ and subset $B \subset [d] \setminus [k + 1]$ is separated by $C = \{k\}$. For $k = 2$:

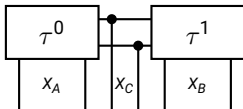


Markov network independencies

Correspondence of separation and independence: When subsets C separates A from B , then we have

$$(X_A \perp X_B) \mid X_C .$$

This can be shown based on the disentanglement of networks (see lecture on CSPs):

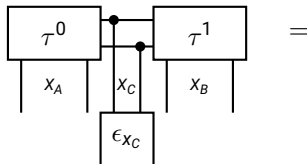


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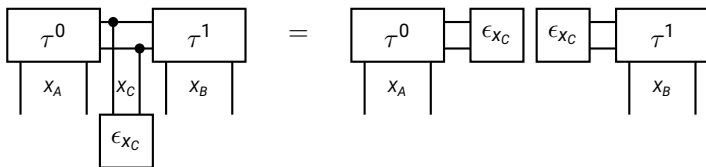


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This can be shown based on the disentanglement of networks (see lecture on CSPs):

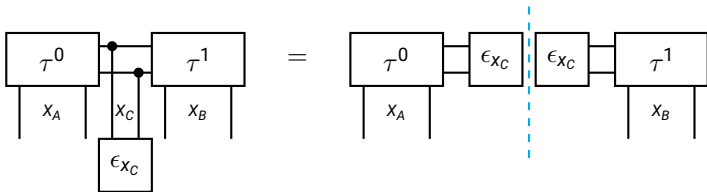


Markov network independencies

Correspondence of separation and independence: When subsets C separates A from B , then we have

$$(X_A \perp X_B) \mid X_C.$$

This can be shown based on the disentanglement of networks (see lecture on CSPs):



Conclusion: Given an assignment to X_C , the variables X_A must be independent of X_B .

Markov network independencies

The other direction is true for positive distributions: We need to assume for any $x_{[d]}$

$$\mathbb{P}[X_{[d]} = x_{[d]}] \neq 0.$$

Theorem (Hammersley Clifford)

A positive distribution $\mathbb{P}[X_{\mathcal{V}}]$ has a representation as a Markov network on a hypergraph $\mathcal{G}$, if and only if for any disjoint subsets $A, B, C \subset \mathcal{V}$ for which C separates A from B

$$(X_A \perp X_B) \mid X_C.$$

Proof: Very technical, see the book [Koller, Friedman: Probabilistic Graphical Models 2009] and alternatively using the contraction formalism the paper [Goessmann et al: A tensor network formalism for neuro-symbolic AI].

Example: Recommending a student

Consider a distribution of the variables:

- ▶ Intelligence of a student: X_I
- ▶ SAT score of the student: X_S
- ▶ Difficulty of a test: X_D
- ▶ Grade of the test: X_G
- ▶ Recommendation by the teacher: X_L

Let us assume the independencies:

- ▶ "The SAT score depends only on the student's intelligence":

$$(X_S \perp X_{\{D,G,L\}}) \mid X_I$$

- ▶ "The recommendation letter depends only on the grade":

$$(X_L \perp X_{\{D,I,S\}}) \mid X_G$$

The corresponding hypergraph:

